

J C Bose University of Science and Technology, YMCA, Faridabad, Deptt of Mech. Engg.

Subject – RAC

Sessional Test-I (25)

MM- 30

Class Vth SM Mech. Engg.

(M51&M52)

Time 90 Min

Attempt any three questions. All questions carry equal marks

1. (a) What are the difference between primary refrigerant and secondary refrigerants? Discuss the desirable properties of refrigerants. (5)

(b) What is the basic working principle of a vapor absorption refrigeration system (VARs)? (5)

2. The following data refer to a reduced ambient refrigeration system :

Ambient pressure = 0.8 bar ,

Pressure of ram air = 1.12 bar

Temperature of ram air = 20 °C,

Pressure at the end of main compressor = 3.33 bar

Efficiency of main compressor = 85% ,

Heat exchanger effectiveness = 80%

Pressure at the exit of the aux. turbine = 0.82 bar, Efficiency of auxiliary turbine = 85%

Temperature of air leaving the cabin = 25°C, Pressure in the cabin = 1.013 bar

Flow rate of air through cabin = 60 kg / min. Find: I. Capacity of the cooling system required; II. Power needed to operate the system; and III. C.O.P. of the system. (10)

3. An ammonia refrigeration machine works between the temperatures of -10°C and 30°C. The vapour leaves the compressor in dry and saturated condition and temperature of the liquid refrigerant leaving the condenser is 30°C. Find the kilograms of ice produced per kW-hour assuming actual COP is 68% of theoretical. The quantity of heat carried per kg of ice is 375 kJ/kg. The properties of ammonia are given below:

(10)

Temp °C	h_f (kJ/kg)	h_{fg} (kJ/kg)	s_f (kJ/kg-K)	s_g (kJ/kg-K)
30	28.5	290.8	0.099	1.055
-10	-8.84	323	-0.033	1.191

4.(a) What is multistage compression? Mention the advantages of it. (5)

(b) Explain the function of each component in a LiBr-based absorption system (Generator, Absorber, Pump, Heat Exchanger, Condenser, Evaporator). (5)